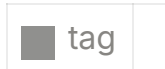


Research Proposal



Ollie

[1. Research Question](#)

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▼ Ollie

Own project: What is the proposed title of your project

From Food Belt to Migration Belt: Assessing the Socio-spatial Impacts of Climate-Induced Agricultural Migration in China

Own project: What is the proposed research question? 50-100 words



Own project: Give a short summary of existing literature on the topic including references (200 words)

Own project: What are the proposed methodologies?

Own project: What is the proposed data? Where will you get it?

Own project: Give a summary of the analysis you plan to carry out (200 words)

Own project: What ethical considerations will you consider within your research (100-200 words)?

1. Research Question

My research asks:

To what extent does climate-induced agricultural disruption drive rural out-migration in China?

Specifically, I aim to understand how climate variability and extreme weather events impact agricultural productivity, and whether these disruptions contribute to rural depopulation patterns across different regions.

2. Related Academic Paper

A key paper that informs my research is:

Liang et al. (2021), Climate-mediated dynamics of the northern limit of paddy rice in China, published in Agricultural and Forest Meteorology.

This study uses remote sensing data to demonstrate how rising temperatures have shifted China's agricultural zones northward, providing a model for how climate impacts agricultural spatial patterns. It supports my approach of using satellite-based vegetation indices to detect agricultural disruption.

3. Overview of Datasets

I will mainly use:

- MODIS NDVI (463m resolution) for agricultural productivity;
- CHIRPS precipitation (0.05°) for rainfall anomalies;
- SPEI drought index (0.5°) for drought severity;
- ERA5-Land temperature anomalies (9km) for heat stress;

- DMSP-OLS and VIIRS night-time lights for rural economic activity and migration proxies;
- County-level population data from Chinese statistical yearbooks for demographic change.

All remote sensing data will be accessed via Google Earth Engine, and processed using Python.

4. Overview of Methods

My research will use a two-tiered spatial-quantitative approach:

- At the national level:
 - I will construct a grid-based panel dataset ($0.5^\circ \times 0.5^\circ$ resolution).
 - I will apply fixed-effects regression models to examine whether inter-annual climate variability impacts vegetation productivity and rural depopulation proxies.
- At the regional level:
 - I will conduct an event-study within a difference-in-differences framework, focusing on counties in Guizhou, Yunnan, or Sichuan impacted by the 2010 Southwest drought.
 - This will explore local mechanisms linking extreme weather, land use decline, and labour migration.

5. Hypothesis

I expect to find that:

- Higher climate variability and increased drought severity will be associated with lower agricultural productivity, as measured by NDVI declines;
- Counties experiencing greater agricultural disruption will show stronger signs of rural out-migration or economic decline, captured by night-time light dimming and population loss.

At the regional level, I hypothesise that areas severely impacted by the 2010 drought will exhibit short-term reductions in land use intensity and accelerated labour outflows compared to unaffected areas.